

MAGNETIC FLUX & FARADAY'S LAW

Magnetic flux $Wb = V \cdot s$

$$\Phi = \iint_S \mathbf{B} \cdot d\mathbf{s}$$

Faraday's law N : number of turns ($N=1$ for single loop)

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} \text{ (V)}$$

Lenz's law: induced current direction opposes the change in Φ .

TRANSFORMER & MOTIONAL EMF

Transformer EMF *stationary loop, time-varying B*

$$V_{\text{emf}}^{\text{tr}} = -N \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *moving conductor in B*

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

Total EMF *general case*

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

$\mathbf{u}$: conductor velocity (m/s). Choose contour direction by *RHR*.

AC GENERATOR

$$V_{\text{emf}} = N A \omega B_0 \sin(\omega t + C_0) \text{ (V)}$$

N : turns; A : loop area (m^2); B_0 : field amp (T); ω : rotation (rad/s).
Derived: $\Phi = N A B_0 \cos(\omega t + C_0)$, $V_{\text{emf}} = -d\Phi/dt$. Peak $V_0 = N A \omega B_0$.

IDEAL TRANSFORMER

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}$$
$$Z_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 Z_L, \quad R_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 R_L$$

Lossless: $V_1 I_1 = V_2 I_2$ (power conserved).
Step-up ($N_2 > N_1$): $V_2 > V_1$, $I_2 < I_1$; step-down: opposite.

MAXWELL'S EQUATIONS

Law	Differential	Integral
Gauss $\mathbf{E}$	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enc}}$
Gauss $\mathbf{B}$	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_c + I_d$

$\mathbf{D} = \epsilon \mathbf{E}$; $\mathbf{B} = \mu \mathbf{H}$; $\mathbf{J} = \sigma \mathbf{E}$ (cond.); $I_c = \iint \mathbf{J} \cdot d\mathbf{s}$; $I_d = \iint \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$.
Phasor: $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$; $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}$; $\epsilon_c = \epsilon' - j\sigma/\omega$.

BOUNDARY CONDITIONS

Field	General	PEC ($\sigma \rightarrow \infty$)
Tang. $\mathbf{E}$	$E_{t1} = E_{t2}$	$E_t = 0$
Tang. $\mathbf{H}$	$H_{t1} - H_{t2} = J_s$	$\hat{n}_2 \times \mathbf{H} = \mathbf{J}_s$
Norm. $\mathbf{D}$	$D_{n1} - D_{n2} = \rho_s$	$D_n = \rho_s$
Norm. $\mathbf{B}$	$B_{n1} = B_{n2}$	$B_n = 0$

J_s : surface current (A/m); ρ_s : surface charge (C/m²); $\hat{n}_2$: normal from 2→1.
No J_s , no ρ_s : E_t , H_t , D_n , B_n all continuous across boundary.

DISPLACEMENT CURRENT

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \text{ (A/m}^2\text{)}, \quad I_d = \iint_S \mathbf{J}_d \cdot d\mathbf{s} \text{ (A)}$$

Phasor: $\tilde{\mathbf{J}}_d = j\omega\epsilon\tilde{\mathbf{E}}$. Completes Ampère; enables EM waves in free space.

CHARGE CONTINUITY & RELAXATION

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \implies \oint_S \mathbf{J} \cdot d\mathbf{s} = 0 \text{ (Kirchhoff's current law)}$$
$$\tau_r = \frac{\epsilon}{\sigma} \text{ (s)}, \quad \rho_v(t) = \rho_{v0} e^{-t/\tau_r} \text{ (C/m}^3\text{)}$$

Good conductor: $\tau_r \approx 10^{-19} \text{ s}$; good insulator: $\tau_r \approx 10^4 \text{ s}$.

LOSSY MEDIUM POYNTING

Average (general) W/m^2

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$$

Lossy medium $\eta_c = |\eta_c| e^{j\theta_\eta}$, wave in $+\hat{z}$

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos\theta_\eta \hat{z} \text{ (W/m}^2\text{)}$$

Lossless $\alpha = 0$, $\theta_\eta = 0$, $|\eta_c| = \eta$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$$

Power $\propto e^{-2\alpha z}$, field $\propto e^{-\alpha z}$: power decays at *twice* the rate of $|\mathbf{E}|$.

ATTENUATION RATE & DECIBELS

Unit conversion

$$\alpha \text{ (dB/m)} = 8.69 \alpha \text{ (Np/m)}$$

Power attenuation rate from $z = 0$

$$A = 10 \log\left(\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)}\right) = -8.69 \alpha z \text{ (dB)}$$

Field attenuation from $z = 0$

$$A = 20 \log\left(\frac{|E(z)|}{|E(0)|}\right) \text{ (dB)}$$

Power/voltage ratio

$$G \text{ (dB)} = 10 \log\left(\frac{P_1}{P_2}\right) = 20 \log\left(\frac{V_1}{V_2}\right)$$

$P \times 2 \Rightarrow +3 \text{ dB}$; $P \times 10 \Rightarrow +10 \text{ dB}$; $P \times 100 \Rightarrow +20 \text{ dB}$.

CURRENT IN GOOD CONDUCTOR

Current density at depth z $J_0 = \sigma E_0$

$$\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \text{ (A/m}^2\text{)}, \quad \delta_s = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (m)}$$

Total current *slab width w (y-dir)*

$$\tilde{I} = \frac{J_0 w \delta_s}{1+j} \text{ (A)}$$

Surface/internal impedance

$$Z_s = \frac{1+j}{\sigma \delta_s} = R_s(1+j), \quad R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \text{ (}\Omega\text{)}$$

AC resistance of slab (width w , length ℓ)

$$R = R_s \frac{\ell}{w} \text{ (}\Omega\text{)}$$

FARADAY QUICK REF

$\Phi = \iint \mathbf{B} \cdot d\mathbf{s}$ (Wb)
 $V_{\text{emf}} = -N d\Phi/dt$
 $V^{\text{tr}} = -N \iint \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$
 $V^{\text{m}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$

TRANSFORMER

$V_1/V_2 = N_1/N_2$
 $I_1/I_2 = N_2/N_1$
 $Z_{\text{in}} = (N_1/N_2)^2 Z_L$
 $V_{\text{emf}} = N A \omega B_0 \sin \omega t$

MAXWELL PHASORS

$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$
 $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}$
 $\epsilon_c = \epsilon' - j\sigma/\omega$
 $\tilde{\mathbf{J}}_d = j\omega\epsilon\tilde{\mathbf{E}}$

dB QUICK REF

$\alpha_{\text{dB}} = 8.69 \alpha_{\text{Np}}$
 $G = 10 \log(P_1/P_2)$
 $G = 20 \log(V_1/V_2)$
 $A = 20 \log(|E(z)|/|E(0)|)$

SURFACE IMPEDANCE

$\delta_s = 1/\sqrt{\pi f \mu \sigma}$ (m)
 $Z_s = (1+j)/(\sigma \delta_s)$
 $R_s = \sqrt{\pi f \mu / \sigma}$ (Ω)
 $R = R_s \ell / w$ (slab, Ω)